DANTE CORSO

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CLOUD INTEGRATION ENGINEER

Engineer who ships event-driven Azure systems that connect telephony, AI services, vendor APIs, and business platforms — and then owns them in production. At a managed services provider, designed, built, and operates four platforms end to end: an AI call-automation pipeline spanning Azure Communication Services, Speech, and Azure OpenAI into Autotask; a multi-vendor network monitor that reconciles detected issues into tickets automatically; and an on-call routing system. Delivers client-facing AI and full-stack products independently across Azure, Firebase, and Google Cloud. Builds for idempotency, traceability, and honest failure reporting under real production load.

TECHNICAL SKILLS

Azure: Functions (Python, Node 20), Communication Services, Event Grid, Speech, Azure OpenAI, AI Foundry, Blob & Table Storage, Static Web Apps, Logic Apps, Key Vault, Managed Identity, Application Insights, Cost Management API

AI Engineering: Azure OpenAI and GPT-4o, Google Gemini, strict JSON-schema output enforcement and re-validation, prompt-injection hardening, batch speech-to-text, OCR (Google Vision), document analysis and summarization pipelines

Integration & APIs: REST API design, JSON Schema validation, OAuth 2.0, webhooks, event-driven architecture, at-least-once delivery semantics, idempotency and correlation tracing; vendor APIs — Autotask PSA, Datto BCDR / SaaS Protection, Axcient x360Recover, UniFi, SonicWall, Cisco Meraki, Microsoft Teams, NVD/EPSS, Stripe

Languages: Python, TypeScript, JavaScript (Node 20), SQL

Web & Data: React 19, Vite, Node.js, Tailwind, TanStack Query, Firebase / Firestore, Cloud Functions, Azure Static Web Apps

Practices & Tooling: Azure DevOps CI/CD, Git, pytest and the Node test runner, dependency-free test doubles, KQL and Application Insights alerting, Key Vault secret management, pure-function design with unit test coverage

PROFESSIONAL EXPERIENCE

E3 IT Services — Cloud Integration Engineer 2024 – Present

Managed IT services provider · Sole engineer on four internal production platforms

Call Automation Platform — ACS → Speech → Azure OpenAI → Autotask

- Architected and operate a production platform across three Python Azure Function Apps that answers inbound Azure Communication Services calls, routes them through an ordered Microsoft Teams engineer queue with automatic voicemail failover, records to Blob Storage, transcribes with Azure Speech batch jobs, and generates client-ready documentation with Azure OpenAI — posting the finished time entry to the engineer's active Autotask ticket. Handles roughly 30 support calls a day (~600/month), removing transcription, summarization, and ticket data entry from the engineer entirely — the call ends and the documentation is already written.
- Designed table-backed idempotency around every external side effect — Event Grid events, call connections, recording IDs, blob uploads, AI analyses, and Autotask posts — so at-least-once webhook delivery and retries cannot produce duplicate recordings, analyses, or ticket notes. Failed operations release their claim row so Event Grid redelivery retries safely.
- Built a sanitized correlation ID propagated across ten hops — ACS callbacks, Logic App notifications, Table partitions, blob metadata, Speech and Azure OpenAI client-request headers, and Autotask idempotency rows — making any call traceable end to end from a single KQL query in Application Insights.
- Hardened the AI layer against prompt injection: transcripts are wrapped as untrusted data and scrubbed of control characters, SAS URLs, connection strings, keys, and instruction-override phrases; output is constrained to a strict JSON schema and re-validated before any external write. Engineer identity resolves from trusted Teams UID metadata, never from model inference.
- Built a React 19 + TypeScript operations dashboard on Azure Static Web Apps with a Node API — service health, storage and table analytics, and a searchable recording and transcript explorer with HTTP Range–based seekable audio — plus Azure Cost Management integration reporting cost per call, with all secrets kept server-side.

Unified Backup System — Datto BCDR / Endpoint / SaaS → Axcient x360Recover

- Built and operate a Node 20 Azure Functions platform — nine functions across three daily timers, three public HTTP endpoints, and three key-protected manual twins — monitoring roughly 525 backup assets on four vendor platforms: 8 BCDR appliances, 224 endpoint agents across 79 clients, 282 SaaS domain/application pairs, and 13 Axcient devices.
- Designed a normalized issue model that collapses every vendor's failures into one record shape of nine types with stable per-issue identifiers, so the same problem carries the same ID every day it exists — which is what makes day-over-day deltas, chronic-failure detection, and one-ticket-per-issue automation possible without duplicates, including correct handling when an issue reclassifies from one type to another.
- Built the system around the invariant that absence of data must never read as absence of problems: per-source status isolates one vendor's outage from the rest, staleness is recomputed at read time rather than trusted from storage, stale caches fail loudly instead of emitting

phantom results, and the state API returns 503 rather than an empty-but-green answer. The rebuild surfaced SaaS monitoring that had been silently dead for nine months and a storage uniqueness key that had been dropping all but roughly 75 of 282 protected pairs.

- Wrote 96 tests on Node's built-in runner with zero test dependencies, substituting the vendor HTTP layer and blob storage in the require cache so collection, normalization, archiving, and reporting run with no cloud, credentials, or network — including outage scenarios impossible to stage in production. The suite gates every deploy through an Azure DevOps pipeline.
- Ships a 7:00 AM Teams exception report in both prose and structured form, an engineer triage console on Azure Static Web Apps refreshing against live endpoints, and an indefinite daily blob archive of raw exports, normalized issues, rollups, and deltas behind a latest-state API that answers in milliseconds without touching a vendor.

Unified Network Monitor — UniFi / SonicWall / Cisco Meraki → Autotask

- Built a multi-vendor monitoring platform on timer-triggered Azure Functions that polls UniFi, SonicWall, and Cisco Meraki APIs every five minutes across roughly 600 endpoints at 100 client sites, normalizes vendor data into Blob snapshots, and reconciles detected issues into Autotask tickets through a Table Storage state machine that opens, touches, and auto-resolves tickets without duplicates.
- Wrote detection as pure, I/O-free Python modules covering site-down, WAN failover, latency, pending-firmware, firewall, VPN, license, and uplink rules, with alert-suppression hierarchies and unit tests across ten modules — collapsing alert storms so one incident produces one ticket. Vendor runs returning inventory-only data skip reconciliation, so a partial API outage cannot auto-resolve live tickets.
- Added a daily advisory pipeline enriching vendor security advisories with NVD CVE detail and FIRST EPSS exploit-probability scores, matching them against deployed firmware and opening tickets automatically. All secrets resolve through Key Vault and managed identity.

On-Call Engineer Routing System

- Built an internal on-call rotation and escalation platform that identifies the assigned engineer for an inbound ticket, sends real-time notifications, and delivers a pre-briefed package — priority, affected systems, and prior history — so the engineer is in context before picking up the phone.

Precision Pixel Innovations — Independent Software Engineer 2023 – Present

Client work delivered end to end: discovery, build, deployment, and ongoing support

- **Funari Public Adjusters — AI Document Analysis Platform.** Built a secure React + Firebase platform for an insurance-adjusting firm. Adjusters upload carrier estimate PDFs; the app runs Google Vision OCR, then uses Gemini to produce side-by-side discrepancy reports that surface underpaid line items, exportable to PDF and Excel. Replaced hours of manual line-by-line comparison with a review that completes in seconds.
- **Dolce Vita Gelato — Full Business Operating System.** Designed, built, and maintain nine integrated systems on one shared real-time Firebase backend: mobile ordering with Stripe checkout and GPS-geofenced arrival alerts, self-order kiosk, digital signage with admin-controlled menus and sold-out toggles, order management, inventory and supply tracking, recipe R&D, and role-gated staff tooling. Live demo: gelato-system-demo.web.app
- **P&S Ravioli Company — Inventory Platform.** Built a real-time inventory platform on Firestore with auto-generated UPC barcodes, multi-location stock tracking, and role-based dashboards for drivers and managers, cutting inventory lookup time by roughly 50%.

EDUCATION & CERTIFICATIONS

B.S. Computer Science, Networking Systems Concentration — Rowan University, May 2024

A.S. Computer Science — Rowan College of South Jersey, 2022

Microsoft AI-900 (2025) · Microsoft MS-900 (2024) · Microsoft AZ-204 in progress · Eagle Scout, BSA